

THE IMPACT OF EXECUTIVE VERBAL COMMUNICATION ON THE CONVERGENCE OF INVESTORS' OPINIONS

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This paper studies the influence of executive verbal communication on the convergence of investors' opinions, defined as reduction of differences in investors' valuations. Building on the corporate communication and sense-giving literatures, we argue that executive verbal communication impacts investor opinion convergence through its influence not only on disclosure of new information, but also on the comprehensibility and credibility of the information conveyed. Accordingly, we hypothesize that the newness, simplicity, and unscriptedness of executive verbal communication are all positively associated with investor opinion convergence. We also argue that the effect of executive communication on investor opinion convergence will be amplified when investors have a greater demand for information as would be the case for those firms with higher earnings uncertainty. Using a unique research design and second-by-second intraday stock trading data from 10,415 conference calls of 847 publicly listed firms between 2005 and 2012, we found strong support for our predictions.

Top executives can influence the financial community in a myriad of ways, including written (e.g., letters to shareholders) and verbal (e.g., earnings conference calls) communication.¹ Recently, a small, but growing, body of work has begun to examine how the attributes of executive verbal communication affect the stock market performance of firms. For example, scholars have found that executives'

use of concrete language (Pan, McNamara, Lee, Halebian, & Devers, 2018) and skillfully framed language (Fiss & Zajac, 2006; Rhee & Fiss, 2014) can influence a firm's stock price positively.

Executive communication can also affect the convergence of investors' opinions. The "opinions" of investors, referring specifically to their valuations of a firm, will converge (or diverge) when differences in their opinions decrease (or increase) (Hong & Stein, 2007; Miller, 1977). When more information is disclosed publicly, the information gap among investors will shrink, and the differences in investor opinions will be reduced. Research has shown that investor opinion convergence is a salient predictor of both a firm's valuation and performance (e.g., Diether, Malloy, & Scherbina, 2002; Doukas, Kim, & Pantzalis, 2006). Importantly, it is an attribute of the financial markets that is especially sensitive to corporate communications (e.g., Berkman, Dimitrov, Jain, Koch, & Tice, 2009; Blankespoor, Miller, & White, 2014). However, the empirical literature on investor opinion convergence is largely confined to written communication (see Lee, 2016, for a notable

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¹ "Verbal communication" refers to communication using words. "Nonverbal communication" refers to communication that does not involve words but relies on the use of gestures, movement, facial expressions, and other means of communication. As such, verbal communication can be oral (spoken) or written. In this paper, however, executive verbal communication corresponds exclusively to oral communication, unless otherwise noted.

exception), and no theory has been advanced of the mechanisms through which verbal communication influences investor opinion convergence.

Addressing this gap and building on the corporate communication and sense-giving literatures, we argue that executive verbal communication impacts investor opinion convergence through its influence not only on the disclosure of new information but also on the perceived comprehensibility and credibility of the information conveyed. According to the sense-giving literature, executive communication can help stakeholders make sense of an ambiguous situation by improving the comprehensibility and credibility of information (Daft & Macintosh, 1981; Petkova, Rindova, & Gupta, 2012). Information “comprehensibility” refers to the ease of understanding a firm’s communications (Eagly, 1974), whereas information “credibility” refers to an audience’s acceptance of a firm’s communications as plausible and trustworthy (Hovland & Weiss, 1951). When a message is hard to understand or highly equivocal, interpretations of that message will likely multiply (Daft & Macintosh, 1981). In parallel, when the trustworthiness of information provided by executives is questionable, investors are less likely to accept and act on it (Hovland & Weiss, 1951). The extent to which executive verbal communication improves the perceived comprehensibility and credibility of the information being conveyed is thus positively associated with the effectiveness of executives’ sense-giving and the convergence of investors’ opinions.

As noted by McCroskey (2001), content, composition, and delivery are three essential aspects of verbal communication. “Content” refers to the message shared with the audience, “composition” to the type of language (i.e., words and expressions) used to craft the message, and “delivery” to the presentation of the message (Den Hartog & Verburg, 1997; Holladay & Coombs, 1993). To explore whether these three essential aspects of verbal communication influence investor opinion convergence, we focused on the *newness* of content (i.e., the distinctiveness of the information provided) (Hanley & Hoberg, 2010), the *simplicity* of composition (i.e., the use of clear and concise language) (Guo, Sengul, & Yu, 2020), and the *unscriptedness* of delivery (i.e., the extent to which a speaker does not rely on a script) (Lee, 2016). We chose these three attributes because they affect the perceived comprehensibility and credibility of information conveyed by executive communication and thus the level of discrepancy in investor opinions. We hypothesized that content newness,

composition simplicity, and delivery unscriptedness of executive verbal communication are all positively associated with investor opinion convergence. We also explored the role of earnings uncertainty as a key contingency that may govern the effect of executive verbal communication on investor opinion convergence. We hypothesized that the future earnings uncertainty of a stock increases investors’ motivation to seek and their willingness to be influenced by additional information, thereby amplifying the positive impact of newness, simplicity, and unscriptedness on investor opinion convergence.

We tested our predictions in the context of earnings conference calls by firms listed on the three major U.S. stock exchanges between January 1, 2005, and May 31, 2012. We examined the effects of executive verbal communication on investor opinion convergence by using second-by-second intraday stock trading data before, during, and after the question-and-answer (Q&A) sessions of 10,415 conference calls. Regression results and robustness checks both strongly supported our predictions.

This study makes three important contributions. First, we innovated by studying all three essential aspects of verbal communication—namely, content, composition, and delivery (McCroskey, 2001)—in tandem. Hence, our paper offers a comprehensive study of the extent to which various aspects of executive verbal communication affect investors’ valuations. Specifically, we theorized and demonstrated that how executives deliver their talks influences investors’ opinions, an aspect of verbal communication largely ignored in the existing research (e.g., König, Mammen, Luger, Fehn, & Enders, 2018; Pan et al., 2018; Whittington, Yakis-Douglas, & Ahn, 2016). Moreover, existing research on opinion convergence has taken a rational view that emphasizes the role of executive communication in supplying more information to investors (e.g., Blankespoor et al., 2014; Lee, 2016). In contrast, we offer a complementary, cognitive perspective that delves into the sense-giving role of executive communication in making information provided more comprehensible and credible.

Second, we have extended the sense-giving literature in two ways. According to Maitlis and Christianson (2014: 81), organizational sense-giving is “more often understood as fundamentally concerned with language.” Most existing work, however, has only focused on the importance of narratives and metaphors as sense-making devices (e.g., Brown, Stacey, & Nandhakumar, 2008; Cornelissen, 2005; Hill & Levenhagen, 1995; Rouleau &

Balogun, 2011). Limited attention has been devoted to the content, delivery, and other characteristics of language (e.g., its simplicity) that we explored in this study. Additionally, the existing work has primarily focused on how executives can use language to pursue support for new ventures (e.g., Lounsbury & Glynn, 2001) and manage strategic change (e.g., Fiss & Zajac, 2006), but paid no attention to information properties of executive communication. Our theory and the findings imply that the effectiveness of executive communication as a sense-giving activity hinges on its effect on perceived comprehensibility and credibility of information to important stakeholders.

Finally, focusing on investor opinion convergence, we have introduced a new and important aspect of investor valuations to the strategy literature. Compared with changes in stock prices, changes in the differences in investors' opinions better depicts investors' sensitivity to firms' information environment and their reactions to corporate communication strategies (e.g., Berkman et al., 2009). To properly measure investor opinion convergence, our use of intraday trading data departs from the studies that have predominantly relied on daily trading data (e.g., Pan et al., 2018; Whittington et al., 2016). Our approach allows us to capture stock market reactions more precisely and limits the influence of potential confounding factors (e.g., Matsumoto, Pronk, & Roelofsen, 2011). Moreover, we used natural language processing techniques to capture content newness and delivery unscriptedness. This represents an early attempt at using a more sophisticated linguistic approach (compared with the word-count approach largely used by existing research) in the strategic management literature.

EXECUTIVE VERBAL COMMUNICATION AND INVESTOR OPINION CONVERGENCE

Many scholars in the field of strategic management have tried to answer the question of how the characteristics and behaviors of top executives affect investors' valuations of a company. A common characteristic of this influential literature has been its focus on one key aspect of investors' valuations—changes in stock prices (see Higgins & Gulati, 2006, for an exception). In principle, a company's stock price reflects its investors' perceptions of the company's health and its prospects for future profitability. Hence, a change in stock price captures the average change in investors' valuations and serves as a good proxy for studying the performance

consequences of a range of corporate decisions such as mergers and acquisitions (e.g., Gaur, Malhotra, & Zhu, 2013), joint ventures or alliance formations (e.g., Madhavan & Prescott, 1995), and governance choices (e.g., Shen & Cannella, 2003).

However, when questions of interest are related to a firm's information environment and the market reactions to its communications (as in our study), a change in the differences in investors' opinions is a more appropriate outcome to explore. After all, more information should reduce the differences in investors' valuations of a firm (because investors are then better able to assess the firm's underlying fundamentals); however, more information may or may not change the stock price. Building on this observation, a body of research in finance and accounting has paid special attention to the convergence or divergence of investors' opinions. For example, these studies reveal that a divergence of opinion among investors can influence post-initial public offering (IPO) stock market performance, takeover premiums, and future stock returns (e.g., Diether et al., 2002; Doukas et al., 2006), and that announcements of corporate earnings increase the convergence of investors' opinions (e.g., Berkman et al., 2009).

Executive Verbal Communication

Within the body of work that links executive communication to investors' decisions, the earlier studies focused primarily on written communication, such as the CEO's letters to shareholders. These revealed that self-interested communications, such as self-serving attributions and the concealment of negative outcomes, helped executives shift the blame from themselves and also helped invoke more favorable responses from the investors (Abrahamson & Park, 1994; Salancik & Meindl, 1984). Only recently have scholars begun to pay close attention to executive verbal communication such as earnings conference calls and strategy presentations (e.g., Shi, Zhang, & Hoskisson, 2019; Whittington et al., 2016).

The literature on executive verbal communication can be assessed based on three essential aspects: content, composition, and delivery (Den Hartog & Verbarg, 1997; McCroskey, 2001). "Content" of verbal communication refers to the message shared with an audience. According to the corporate disclosure literature, the amount of information content in executive communication shapes the extent to which investors are informed (Healy & Palepu, 2001). However, not all content is equal in terms of its information value. It is well documented that

executives sometimes view disclosure mainly as a means to satisfy regulatory demands, rather than to communicate information to investors (e.g., Brown & Tucker, 2011; Hanley & Hoberg, 2010; Lang & Stice-Lawrence, 2015). Therefore, regulators and researchers have frequently cited the prevalence of a so-called “boilerplate” content as a concern because it offers little real information in executive communication (e.g., Hoogervorst, 2013). Accordingly, it is necessary to differentiate informative content from uninformative content. For instance, decomposing the content of a firm’s initial IPO prospectus into “standard content” (i.e., duplicated information contained in past industry IPOs) and “informative content” (i.e., new and unique information in the focal IPO), Hanley and Hoberg (2010) found that a greater amount of informative content in an IPO prospectus decreases underpricing (a key measure of information asymmetry in an IPO setting), while an increase in the amount of standard content in that prospectus has the opposite effect.²

The “composition” of verbal communication refers to the type of language (i.e., words and expressions) used to craft the messages (Den Hartog & Verbarg, 1997). A growing amount of evidence indicates that even messages with nearly identical content can actually have drastically different impacts depending on the type of language used to construct them (Pennebaker, Mehl, & Niederhoffer, 2003). Prior research in this area has examined the influence of several types of language, such as metaphorical (e.g., König et al., 2018), vivid (Emrich, Brower, Feldman, & Garland, 2001) and complex language (e.g., Crilly, Hansen, & Zollo, 2016; Guo et al., 2020). For instance, Pan and colleagues (2018) showed that investors react favorably to executives’ use of concrete language in verbal communication because concrete language helped executives to be perceived as more competent. Of all the attributes of composition, complexity has been most widely studied.

Finally, “delivery” refers to the actual presentation of the message (Holladay & Coombs, 1993). Unlike the content and composition of executive communication, we know little about whether and how delivery matters in financial markets. In

experimental settings, several leadership studies in organizational behavior have found that a leader’s verbal and nonverbal delivery (e.g., body cues) influences the perceptions of the leader’s charisma and effectiveness (e.g., Awamleh & Gardner, 1999; Holladay & Coombs, 1994). Whereas most of the research on nonverbal delivery has focused on the use of body cues, such as gestures (e.g., Cornelissen, Clarke, & Cienki, 2012), movement (e.g., Gundersen & Hopper, 1976), eye contact (e.g., Beebe, 1976), and facial expressions (e.g., Choudhury, Wang, Carlson, & Khanna, 2019), most of the research on verbal delivery has focused on the use of voice, such as rate (e.g., Filippi et al., 2017), volume (e.g., Gundersen & Hopper, 1976), pitch (e.g., Mayew & Venkatachalam, 2012), and pauses (e.g., Dufour, Estève, & Deléglise, 2014). Building on the observation that successful delivery needs to be natural and conversational (Flemming, 1920; Parrish, 1951), a small group of researchers examined one aspect of verbal delivery that could be studied using text analysis (without any audiovisual data); namely, the use of a script, or the extent to which a talk is delivered by adhering to a predetermined script (e.g., Hildebrandt & Stevens, 1963; Lee, 2016). For instance, Lee (2016) found that, when executives respond to questions from prepared scripts during conference calls, the stock market tends to react negatively.

The Effect of Executive Verbal Communication on Investor Opinion Convergence

One important way to reduce the differences of opinion across investors is to release more information publicly to lessen the information disadvantage of less informed investors. Investors need timely information about a firm’s actions and performance in order to produce sensible investment valuations. As underscored by Arthur Levitt, a former U.S. Securities and Exchange Commission (SEC) chair, information is “the lifeblood of strong, vibrant markets” (Levitt, 1999). It is also widely recognized that investors are diversely informed (e.g., Grossman, 1976). Although some investors are well informed about the operations of a firm and its management, others have access only to limited and imprecise information about that firm. As a result, well-informed investors tend to value a firm’s stock at a price closer to its fundamental value, but investors with less information tend to value the same firm either too high or too low. Therefore, when more

² Note that, even though the influence of the content of executive communication is intuitive and frequently discussed in the literature and in practice, the relevant empirical evidence found in the corporate disclosure literature comes exclusively from studies of written communication (i.e., IPO prospectuses and 10-K reports), leaving the content of verbal communication relatively unexplored.

information about the firm is released publicly, the information gap among investors shrinks.

An alternative to reduce the differences of opinion across investors is to align their divergent interpretations of organizational reality through sense-giving. "Sense-giving" is a process by which people endow their speech with meaning (Polanyi, 1967) as they try "to influence others' understanding of an issue" (Maitlis, 2005: 21) and focus their attention on selected organizational activities (Petkova et al., 2012). Prior research suggests that the most powerful impact of managers is "in shaping the interpretations of others through symbolic language and action" (Lant, 2002: 352; Pfeffer, 1981; Smircich & Stubbart, 1985). As sense-givers (Foldy, Goldman, & Ospina, 2008; Maitlis & Lawrence, 2007; Rouleau, 2005), managers provide explanations, rationalizations, and legitimation for the actions undertaken by their organizations through discursive construction of reality (Antaki, 1994). For example, in the context of new ventures, research has shown that executives' use of narratives and metaphors enable them to explain their vision (Santos & Eisenhardt, 2009), intended growth strategy (Porac, Mishina, & Pollock, 2002), market aspirations (Martens, Jennings, & Jennings, 2007) and resources (Hill & Levenhagen, 1995) to current and prospective investors. Moreover, in the context of strategic change, research has demonstrated that leaders can tailor their narratives in ways that appeal to the key stakeholders to influence how they make sense of the intended change (e.g., Fiss & Zajac, 2006; Sonenshein, 2010). Although the decisions and actions of a firm are objective facts, the interpretation of their meanings is often socially constructed (Weick, 1995). Therefore, in the sense-giving literature, language is not a mere reflection of organizational reality but a powerful tool with which to shape how others perceive and understand an organization (Gioia & Chittipeddi, 1991; Maitlis & Christianson, 2014). As such, executive communication can be seen as a sense-giving activity through which executives attempt to "influence the sense-making and meaning construction of others toward a preferred redefinition of organizational reality" (Gioia & Chittipeddi, 1991: 442).

Following prior research (Petkova et al., 2012), we expect that the effectiveness of executive communication as sense-giving activities hinges on the properties of the information conveyed. Specifically, we emphasize the importance of the perceived comprehensibility and credibility of information (Daft & Macintosh,

1981; Petkova et al., 2012) in helping stakeholders make sense of an ambiguous situation. "Information comprehensibility" relates to the ease of understanding a firm's communication (Daft & Macintosh, 1981; Eagly, 1974). "Information credibility" is the extent to which an audience accepts a firm's communication as both plausible and trustworthy (Hovland & Weiss, 1951). When a message being conveyed is hard to understand or highly equivocal, different interpretations of it multiply (Daft & Macintosh, 1981). In parallel, when the trustworthiness of the information provided by executives is questionable, investors are less likely to accept and act on it (Hovland & Weiss, 1951; Petkova et al., 2012). Hence, the greater the extent to which executive verbal communication improves the perceived comprehensibility and credibility of the information conveyed, the more such communication can help a firm's investors make sense of organizational activities; accordingly, the differences in investors' valuations of the firm should be expected to decrease.

Applying the sense-giving perspective, we propose that the newness, simplicity, and unscriptedness of executive verbal communication are positively associated with investor opinion convergence. This is because these three features increase the perceived comprehensibility and credibility of the information being conveyed, thus improving the effectiveness of executive communication as a tool for sense-giving, and hence reducing the information gap between investors.

Content newness. The extent to which executive communication reveals new information is associated with an increase in the level of convergence of investors' opinions for several reasons. First, and most directly, an increase in the amount of available information reduces the information gap between diversely informed investors (Blankespoor et al., 2014; Van Buskirk, 2012). As a result, the level of discrepancy in those investors' evaluations decreases, and thereby the evaluation of the firm by investors as a whole converges. Second, new content increases perceived comprehensibility because the attention it attracts is a precondition for better comprehension (Petkova et al., 2012). This is especially the case when the newly released content is distinct from what has been previously disclosed (Brown & Tucker, 2011; Hanley & Hoberg, 2010). Third, new content may enhance the perceived credibility of executive communication. The completeness of coverage is an important determinant of the perceived

quality of information (Rieh & Belkin, 1998), which increases the credibility of the communication in the eyes of the audience (Hamilton, 1998; Yang, Kang, & Johnson, 2010). As the completeness of coverage is positively associated with the amount of new content, investors would perceive executives' talk with more new content to be more credible. Taken together, executives' ability to influence the sense-making of investors depends on the perceived comprehensibility and credibility of their communication. We argue that the level of new content in executive communication increases the amount of information provided, augments the perceived comprehensibility and credibility of the available information, and hence is associated with a higher convergence of investors' opinions. Thus, we posit:

Hypothesis 1. The level of content newness in executive communication is positively associated with investor opinion convergence.

Composition simplicity. "Simplicity" of language refers to the use of clear and concise expressions (Kimble, 2012). Simple communication avoids the use of obscure words, unduly complex words, and confusing jargon, such as "scalable," "interoperability," and "groundbreaking." It also uses shorter phrases rather than longer ones (Pallotti, 2015). Because of the value of linguistic simplicity for learning and comprehension, it has become a widely endorsed communication practice by governments (SEC, 1998), legal professionals (e.g., Garner, 2013), educators (e.g., Pallotti, 2015), and researchers (e.g., Miller, 2010).

Composition simplicity increases the convergence of investors' opinions in two important ways. First, it makes it easier for investors to make sense of and remember the messages delivered in the executive communications. In general, longer words and sentences place a greater burden on message recipients because they have to retain more ideas before deciding on the full meaning of the message (Chaiken & Eagly, 1976; DuBay, 2004). Moreover, a higher level of formal education is required to comprehend information communicated by using complex words and convoluted expressions (DuBay, 2004). Composition simplicity improves comprehension because it reduces the burden on memory and eliminates learning barriers for less-educated investors. Consistent with this view, research has found that composition simplicity is a strong predictor of message recall (Chaiken & Eagly, 1976; McGuire, 1968). Second, composition simplicity enhances the credibility and the sense-giving role of executive communication. It

has been shown that individuals are more likely to rely on easily processed information (e.g., easy-to-pronounce words) for decision-making (see Alter & Oppenheimer, 2009, for a review). Prior research in psychology and marketing also suggests individuals tend to view easily processed information as more reliable and more credible (e.g., Rennekamp, 2012; Shah, 2007). In sum, we argue that the use of simple language in executive communication will improve the perceived comprehensibility and credibility of the information being conveyed and thus increase the convergence of investor opinions. Hence:

Hypothesis 2. The level of composition simplicity in executive communication is positively associated with investor opinion convergence.

Delivery unscriptedness. "Delivery unscriptedness" relates to the extent to which a speaker delivers a talk without a script (Lee, 2016). Although a scripted talk can offer many advantages, such as maximum precision of ideas and well-crafted language, in most cases, such talks will be perceived as overly mechanical and unengaging (Lucas, 2009). In contrast, an unscripted talk gives the speaker flexibility to adapt the content and language according to a specific audience and its members' reactions. It is, by definition, more spontaneous because the exact wording appears to be chosen at the time of delivery (Buck & VanLear, 2002; Dufour et al., 2014). However, a less scripted talk also increases the risk that a speaker may misspeak or release key information inadvertently (Lee, 2016).

We argue that the unscriptedness of executive communication increases the convergence of investors' opinions for two reasons. First, such communications are more comprehensible and facilitate executives' sense-giving efforts to shape investors' perceptions of organizational activities. Written and spoken language can differ significantly in terms of structure, content density, and word usage (Chaiken & Eagly, 1976; McCarthy & Slade, 2007). Structurally, written language is more complex than spoken language, uses longer words and sentences, and tends to convey more detailed information in each sentence. As a result, compared with spoken language, written language requires more cognitive effort to process and comprehend. Because unscripted communication is typically delivered conversationally, it uses more spoken language than written language (Lucas, 2009; Lundell, 2009), thereby making it easier for the audience to comprehend.

Second, unscripted communications enhance the sense-giving role of executive communication by boosting its credibility. Because perceived authenticity is closely related to openness and straightforwardness (Molleda, 2010; Yang et al., 2010), communication that suggests openness and straightforwardness makes executives seem more authentic and more credible. Indeed, communicating via unscripted and unpolished statements—in contrast to reading directly from a script prepared by the legal and investor relations teams of a company—may give an audience the impression that executives are willing to be more open and forthright (Clark & Wasow, 1998; Yang et al., 2010). As a result, investors are more likely to perceive such unscripted communication as credible. In sum, we argue that delivery unscriptedness improves the perceived comprehensibility and credibility of information, enhances the effectiveness of executives' sense-giving, and thus increases the level of convergence of investors' opinions. Thus:

Hypothesis 3. The level of delivery unscriptedness in executive communication is positively associated with investor opinion convergence.

The Moderating Role of Earnings Uncertainty

We have argued in the previous section that the levels of newness, simplicity, and unscriptedness of executive communication are all associated with greater convergence of investors' opinions. In this section, we further argue that the effect of communication on investor opinion convergence will be amplified for those firms with high earnings uncertainty. "Earnings uncertainty" refers to ambiguity about future earnings that will accrue to the stockholders of a firm. It also can be thought of as uncertainty about the stockholders' earnings tomorrow per dollars invested in the stock of a firm today (Chen & Zhang, 1998). Because of firm and industry heterogeneities, firms differ in terms of their prospects in the eyes of investors. For example, future earnings uncertainty is substantially greater for firms under distress and/or those that have high financial leverage (Fama & French, 1995; Griffin & Lemmon, 2002).

For firms with high earnings uncertainty, investors' demands for information and their susceptibility to it are both high because accurately evaluating these companies and predicting their future returns becomes daunting (Bauguess, Cooney, & Hanley, 2018). Accordingly, the heightened uncertainty of a

firm will make the sense-giving role of its executive communication much more salient (Petkova et al., 2012). Because investors' attention and scrutiny of executive communication will be enhanced when their demands for information and their openness to it are higher, we predict that the levels of newness, simplicity, and unscriptedness of executive verbal communication will have a stronger effect on the investor opinion convergence when a firm has a high degree of future earnings uncertainty. Thus, we postulate:

Hypothesis 4. The higher the focal firm's earnings uncertainty, the stronger the positive relationship between the level of (a) content newness, (b) composition simplicity, and (c) delivery unscriptedness in executive communication and investor opinion convergence.

METHODOLOGY

Sample

We used data on earnings conference calls from firms listed on the former American Stock Exchange (AMEX; later NYSE American), the Nasdaq, and the New York Stock Exchange (NYSE) between January 1, 2005, and May 31, 2012, to test our theory. This data set came from five sources. We gathered per-second intraday stock trading data from the NYSE Trade and Quote (TAQ) database; historical trading data from the Center for Research in Security Prices (CRSP); and transcripts of corporate earnings conference calls from the Refinitiv's Company Events Coverage database (formerly known as Thomson Reuters StreetEvents); annual and quarterly financial data from Compustat; and firms' quarterly earnings and analysts' consensus forecasts from the Institutional Broker Estimate System (I/B/E/S).

We constructed our sample using four criteria. First, we restricted our sample to conference calls held between 11 a.m. and 2 p.m. Eastern Standard Time (EST) (all three stock exchanges are in New York). These cutoffs were necessary because our research design required us to observe stock market reactions immediately before, during, and immediately after the call (the stock market opens at 9:30 a.m. and closes at 4 p.m.). Limiting the sample to trading hours is a standard practice in studies that use intraday stock trading data (e.g., Frankel, Johnson, & Skinner, 1999; Matsumoto et al., 2011), because it ensures that trading activities can be observed around specific events, thereby

minimizing the influence of other potentially confounding events.

Second, we restricted our sample to conference calls held shortly after earnings announcements. Accordingly, as in prior studies (e.g., Chen, Demers, & Lev, 2018), we excluded calls that were held more than two days after an earnings announcement. Third, we dropped all calls that were held less than 90 minutes after the earnings announcement, as in Frankel et al. (1999).³ This decision ensured that our pre-call measurement of the differences in investors' opinions would factor in the information that was provided in the announcement in such a way that our dependent variable (which compares differences in investors' opinions before and after the call) could capture the effect of the call and not the announcement. Finally, we eliminated those conference calls in which neither the management discussion nor the Q&A session exceeded 200 words. This process allowed us to screen out conference calls that may not have been open to the public and hence were unsuitable for our study. After we applied these criteria and eliminated observations with missing data, our final working sample contained 10,415 observations for 847 unique firms operating in 237 four-digit Standard Industrial Classification (SIC) industries.

Research Design

To examine the effect of executive communication on investor opinion convergence, we studied the Q&A sessions in conference calls that came after earnings announcements. Most public firms hold these calls shortly after their earnings announcements to help market participants make sense of their recently announced earnings and provide their own outlook for the future. Conference calls are accessible to virtually everyone, and they can be a vital source of information to company outsiders. In recent years, conference calls have become a standard method used by publicly traded firms to communicate with their investors (Frankel et al., 1999).

A conference call has two distinct components; namely, a management discussion (MD) and a Q&A session. The MD portion of a call is usually an uninterrupted PowerPoint presentation delivered by top executives. Many top executives commonly refer to the MDs as "prepared remarks" or "prepared comments" (Lee, 2016) because they are typically

based on written scripts prepared by a firm's investor relations and legal teams. The Q&A portion of the call is an interactive session in which participants, such as financial analysts, ask executives questions. Top executives are expected to address these participants' questions and comment on the spot (Hollander, Pronk, & Roelofsen, 2010). According to the 2014 Earnings Call Practices Survey conducted by the National Investor Relations Institute (NIRI), a professional association of corporate officers and investor relations consultants, 81% of these companies script answers to potential questions that could appear during the Q&A. The MD typically lasts for 15–20 minutes, followed by a Q&A session that usually lasts for another 30 to 45 minutes (Frankel et al., 1999).

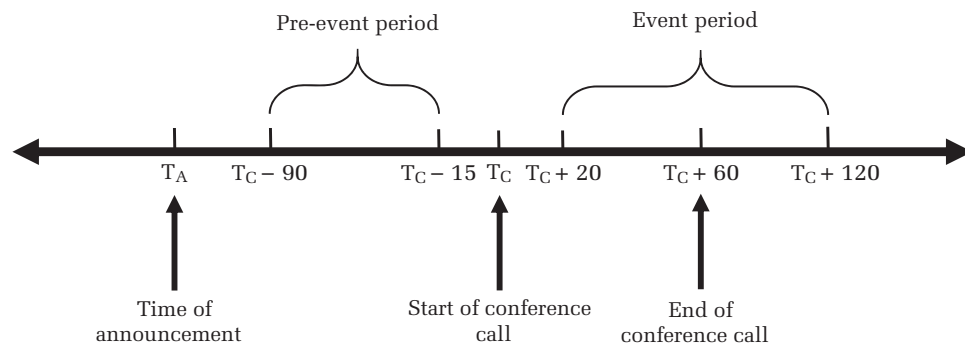
To empirically examine whether executives' verbal responses to the questions during the Q&A (the event) influence investor opinion convergence, we compared differences in investors' opinions during the pre-event period to differences in investors' opinions during the event period. In line with the prior research (e.g., Matsumoto et al., 2011), we defined a "pre-event period" as one that began 90 minutes before the scheduled start time of the call and ended 15 minutes before the start time of the call; we defined the "event period" as a period that begins 20 minutes after the scheduled start time of the call—the typical start time for a Q&A session—and ends 100 minutes later (i.e., about one hour after the end of the call's Q&A session). Hence, the event period captures both real-time trading by investors during the 40 minutes of Q&A time and one hour after the event.

Substantial evidence indicates that investors execute trades in real time during corporate news events, and price adjustments will typically take place within an hour (Busse & Clifton Green, 2002). Accordingly, we gathered per-second trading data for both the pre-event and event periods. With respect to the 10,415 calls in our working sample, this approach amounted to about 110 million seconds of trading data that we used to calculate investor opinion convergence for each call, as we described below.

As outlined in Figure 1, our research design presented the following sequence of events: (1) the company makes an earnings announcement (T_A); (2) the market reacts to the announcement and differences in investors' opinions are captured for the period just before the conference call (between $T_C - 90$ minutes and $T_C - 15$ minutes); (3) the conference call starts with a MD (T_C); (4) the Q&A session of the

³ Most of the firms in our sample (94%) scheduled their conference calls at least 90 minutes after they issued their earnings releases.

FIGURE 1
Timeline for Earnings Announcement, Pre-Event Period, and Event Period (Measured in Minutes)



call starts ($T_C + 20$ minutes), and the call ends 40 minutes later ($T_C + 60$ minutes); and (5) the differences in investors' opinions are captured again for the period between the start of the Q&A session ($T_C + 20$ minutes) and one hour after the end of the call (two hours after the start of the call, $T_C + 120$ minutes). It is important to emphasize that, in our data, the pre-event period always occurs after the earnings announcement. This choice is important in the research design because the information that comes from the earnings announcement is already incorporated into the pre-event valuation of the investors.

Compared with daily data, intraday trading data allowed us to capture stock market reactions with more precision, for two main reasons. First, the liquidity of equity markets has become much faster since the beginning of the 21st century (Holden & Jacobsen, 2014). For example, Chordia, Roll, and Subrahmanyam (2011) reported a 33-fold increase in the frequency of trades in NYSE stocks between January 2003 and June 2008.⁴ This change implies that the potential influence of confounding events in the daily analyses of trading activity has increased, thereby calling for a more fine-grained level of data to accurately measure stock market reactions.⁵

⁴ This increase in liquidity reflects in part an increase in computer-aided trading. Mechanically, increased liquidity means a narrower bid-ask spread because traders can execute trade orders with more information and at a better average price in a more liquid market (Holden & Jacobsen, 2014). Hence, as bid-ask spreads become narrower in response to the increased use of computers, it becomes harder to find support for our arguments.

⁵ Most prior studies that have examined how investors react to corporate communication (e.g., Pan et al., 2018;

Second, investors execute trades in real time during corporate news events (e.g., Busse & Clifton Green, 2002). Thus, stock market reactions to earnings announcements and conference calls will occur within minutes after these events. Consequently, teasing out investors' reactions to a single event becomes a daunting task when multiple events on the same day are likely to introduce noise into the investors' daily reactions. As a result, using intraday trading data, as we did here, has become the state-of-the-art approach in accounting and finance studies for assessing investors' reactions to critical events (e.g., Chen et al., 2018; Matsumoto et al., 2011).

Dependent Variable: Investor Opinion Convergence

Consistent with the corresponding literature in accounting and finance (e.g., Blankespoor et al., 2014; Collier & Yohn, 2006; Doukas et al., 2006; McNish & Wood, 1992), we used bid-ask spreads to capture the differences in investors' opinions. A "bid-ask spread" is the amount by which the best ask price (quoted offer price) exceeds the best bid price (quoted bid price) for an asset in the market, and it is a good proxy for investors' differential access to information. Thus, a reduction in the bid-ask spread is an indication that the valuation differences among the investors have decreased.⁶

Whittington et al., 2016) have aggregated investors' reactions over a two- or three-day window—typically, the day of the event and the day after the event.

⁶ When the information gap between the investor declines, the bid-ask spread is expected to decrease, and trading volume is expected to increase (Garfinkel &

Accordingly, when the bid–ask spread has decreased after a conference call compared with the pre-call level, we interpret it as a convergence of the investors' opinions.

More specifically, to construct this variable, we first calculated the pre-event period bid–ask spread as the average per-second difference between the highest quoted bid and the highest quoted offer price, divided by the midpoint between those two prices, during that time period as follows:

$Spread_{pre-event\ period}$

$$= \sum_{s=1}^n \frac{Best\ ask_s - Best\ bid_s}{Midpoint\ between\ ask\ and\ bid_s} \times 100$$

where s is a given second of the period, ranging between 1 (the first second of the period) and n (the last second of the period). We calculated the event period bid–ask spread in the same way:

$Spread_{event\ period}$

$$= \sum_{s=1}^n \frac{Best\ ask_s - Best\ bid_s}{Midpoint\ between\ ask\ and\ bid_s} \times 100$$

We then calculated *investor opinion convergence* as the difference in the bid–ask spread between these two periods:

Investor Opinion Convergence

$$= Spread_{pre-event\ period} - Spread_{event\ period}$$

When this value is positive, it indicates that the differences of opinion among investors have decreased (i.e., the investors' opinions have converged) after the Q&A session for the earnings call. When this value is negative, it indicates that the differences of opinion have increased (i.e., the investors' opinions have instead diverged) after the Q&A session.

Note that our dependent variable is a difference score, the use of which has been the subject of a

heated debate in the social sciences (e.g., Johns, 1981; Rogosa & Willett, 1983; Tisak & Smith, 1994). Nevertheless, we believe our dependent variable is suitable for this study, for three reasons. First, we studied differences in the same construct (i.e., investor opinion spread) measured at two different points in time. This is a special type of difference score that is often referred to as a “change score” or “gain score” (Lewis-Beck, Bryman, & Liao, 2004). Unlike other types of difference scores (such as fit or congruence), change scores are a powerful measure of change and are highly robust under many situations, including the presence of measurement errors (Allison, 1990; Edwards, 2001). Second, we compared the differences in investors' opinions between the event period and the pre-event period, as in the prior work (e.g., Chatterjee, John, & Yan, 2012; Doukas et al., 2006), rather than using the value of differences in investors' opinions after the Q&A session ($spread_{event\ period}$) as the dependent variable. We adopted this approach because it is challenging to interpret the magnitude of differences in investors' opinions after the Q&A session without a benchmark. Moreover, such a design cleanly captures the influence of executives' verbal responses in the Q&A by mitigating the noise introduced by the earnings announcements and other economic and firm-specific events that occurred before the conference call. Finally, using a well-established measure helps us increase the validity and comparability of our analysis. As we noted earlier, changes in the bid–ask spread is the most commonly used measure of investor opinion convergence/divergence in the literature.

Independent Variables

Content newness. To capture the extent to which executives' responses in a Q&A included new content unmentioned in the MD, we followed the prior research (e.g., Brown & Tucker, 2011; Lee, 2016) and examined the differences in executives' use of content words across these two portions of the conference call. Content words are the nouns, regular verbs, adjectives, and adverbs in a sentence that convey meaning (Chung & Pennebaker, 2007). For example, in the following sentence, “Obama speaks to the media in Oregon,” the content words are “Obama,” “speaks,” “media,” and “Oregon,” while the function words that glue the content words together are “to,” “the,” and “in.”

When comparing the content of two text documents, we considered two types of similarities: “lexical similarity” (which captures the number of

Sokobin, 2006). Accordingly, in the supplementary regressions, we replicated our analyses using as the dependent variable the *abnormal trading volume* (measured as the total number of shares traded divided by the number of shares outstanding on the day after the conference call, adjusted for the trading of stocks at the market level that day and the typical trading volume for a firm's stock over the last 50 days). As expected, our independent variables were positively and significantly associated with a firm's abnormal trading volume ($p < .01$).

words in common) and “semantic similarity” (which captures the similarity of the meanings of those words). It is important to take into account both types of similarities because understanding one type without the other may be misleading. Take for example, the two following sentences: “Obama speaks to the media in Oregon” and “The President greets the press in Portland.” Because these sentences have no common words, the two texts will have a lexical similarity of 0. However, these sentences still convey nearly identical information. This problem can be solved by taking semantic similarity into account by using large-scale pre-trained models to map the meaning of the words into a word-embedding space to capture their closeness. Returning to the example above, after considering the semantic similarity in the meanings of all pairs of words for these two sentences, the content similarity between these two sentences increases to 81.25%. This increase results from taking into account several pairs of words that have a very high semantic similarity score in the text-embedding space; namely, “Obama” and “President,” “speaks” and “greets,” “media” and “press,” and “Oregon” and “Portland.”

Consequently, researchers are increasingly recommended to use a combination of “cosine similarity” (which accounts for lexical similarity) and “text embedding” (which accounts for semantic similarity) to compare the contents of two documents (e.g., Baroni, Dinu, & Kruszewski, 2014; Jurafsky & Martin, 2008). In line with this recommendation, we created our measure of content newness by using the following procedure. First, we created two vectors of the counts of all content words spoken by executives in the MD and Q&A. Second, we took these content words and used text-embedding vectors (composed of about 100 billion words from the Google News data set) to identify the semantic similarity of the meanings of each pair of content words.⁷ This semantic similarity score between two content words ranged between 0 (indicating that the two words have no common meaning) and 1 (indicating that the two words have an identical meanings).

Third, we calculated the dot product of the two vectors we created in the first step for each word, weighing each word pair by the semantic similarity scores created in the second step. We then normalized the calculated score by the length of these two vectors. Note that this process differs from the standard cosine similarity measure, which takes into account only the words common to both documents (which, by definition, would have a semantic similarity score of 1). Finally, we reverse-coded this measure to capture the content newness of the executive communication and adjusted it for document length according to the procedure used by Lee (2016).⁸ Although the cosine similarity approach normalizes for document length, these adjustments are necessary because “longer documents allow for a greater probability that a word is included in both documents, leading to mechanically greater similarity between them” (Lee, 2016: 233).

To assess the validity of our measure, we extracted the number of questions asked by analysts during the Q&A sessions of conference calls. Prior research suggests that analysts’ questions prompt executives to clarify points mentioned in the MD and/or discuss points not already mentioned (e.g., Hollander et al., 2010). Hence, it stands to reason that the amount of information revealed in the Q&A portion of conference calls increases with the number of questions asked by analysts. We found that the correlation between our measure of content newness and the number of questions asked by analysts was positive and highly significant ($r = .10$, $p < .001$). We interpreted this result as an indication of convergent validity for our measure.

Composition simplicity. To capture the extent to which executives used simple and concise language in their verbal communication, we started by calculating the “Fog index” of the Q&As. The Fog index is one of the most widely used measures of language complexity (e.g., Bushee, Gow, & Taylor, 2018; Guo et al., 2020; Miller, 2010). It captures two important aspects of language complexity; namely, “sentence

⁷ For word embedding, we used Word2Vec, a widely used model created by Google engineers (Mikolov, Chen, Corrado, & Dean, 2013). This model is not only computationally efficient but also generates word embeddings of high quality (e.g., Baroni et al., 2014). In a supplementary analysis, the use of a topic modeling approach, Latent Dirichlet Allocation (LDA; see Huang, Lehav, Zang, & Zheng, 2018), instead of Word2Vec yielded somewhat weaker, but still consistent, results.

⁸ We first estimated the relationship between content newness and the length of the MD and Q&A documents and then used the residual of this regression, which is a length-adjusted measure of content newness to predict opinion convergence. An alternative method used in computational linguistics for document length adjustment is TF-IDF (term frequency-inverse document frequency; Loughran & McDonald, 2016). Supplementary regressions using this method, instead of the one used by Lee (2016), yielded qualitatively identical results.

length” (i.e., the average length of sentences being used) and “word complexity” (i.e., the number of words containing three or more syllables). Specifically, the Fog index is calculated as follows:

$$\text{Fog Index} = 0.4 \times \left[\left(\frac{\text{total words}}{\text{total sentences}} \right) + 100 \times \left(\frac{\text{complex words}}{\text{total words}} \right) \right]$$

This score indicates how many years of formal education a person needs to understand a message. For example, a Fog score of 12 means that the reading level of a U.S. high school senior will be required to understand that message on the first reading. Prior research has shown that, if a message needs to be understood by a wide audience, its complexity level should not exceed a Fog score of 8, which means the reading level of an eighth-grade student in the United States. If a message scores a Fog index of 18 or higher, it is incomprehensible for the general public. The mean of language complexity in our sample was 15.42, meaning that a typical conference call would be appropriate for a junior in college in the United States to understand.⁹ About 1% of the conference calls in our sample received a Fog score of over 20, which means those conference calls were close to incomprehensible for the general public. We reverse-coded this measure by subtracting it from 24 (which equals to the maximum Fog score in our working sample plus 1) to capture the composition simplicity of executives’ responses during the Q&As.

To assess the validity of our measure, we first correlated it with another compositional characteristic that may also increase information comprehensiveness and credibility: “language clarity” (i.e., the use of clear and unequivocal language) (Guo et al., 2020). We found a positive and highly significant correlation between these two measures (.15, $p < .001$). Because these measures are conceptually similar but measured by different methods, we considered their strong correlation as evidence of the

convergent validity of our measure. Next, to check the divergent validity of composition simplicity, we checked the correlation between our measure and three compositional characteristics that are conceptually unrelated to composition simplicity: “language activity” (the use of language that indicates movement, action, change, and the implementation of ideas) (Seyranian & Bligh, 2008), “certainty” (the use of language that indicates resoluteness and inflexibility) (Graf-Vlachy, Bundy, & Hambrick, 2020), and “intensity” (the use of language that indicates strength, extremity, and emotionalism) (Hamilton & Hunter, 1998). The correlations between composition simplicity and language activity, certainty, and intensity are $-.002$, $-.01$, and $-.002$, respectively, and none of these correlations is significant.¹⁰

Delivery unscriptedness. The communication literature provides considerable evidence that the verbal delivery of scripted and unscripted communications will differ fundamentally (e.g., Clark & Wasow, 1998; Ward, 1991). Thus, how the Q&A portion of an earnings conference call (which is often partially unscripted) is delivered could be substantially different from how the MD portion (the scripted communication) is delivered. To capture the degree to which executives deliver the Q&A session by going off the script used by the MD that precedes it, we focused on the differences in executives’ use of function words between the two portions of a conference call. Function words—such as pronouns, articles, prepositions, conjunctives, and auxiliary words—primarily serve grammatical purposes and do not have meaningful content (Chung & Pennebaker, 2007). Although function words represent a relatively small component of our vocabulary (less than 0.04%), they account for more than half of the words we use in daily communication. Research in computational linguistics has shown that function words are the best discriminators at capturing the linguistic style of a message (i.e., how the speaker says something) (e.g., Koppel, Schler, & Argamon, 2009; Stamatatos, 2009) because they are subject to less conscious control by the speaker than content words (Chung & Pennebaker, 2007).

Like Lee (2016), we used *cosine similarity* to calculate the level of similarity in executives’ use of function words between the MD and Q&A portions

⁹ The standard deviation of language complexity (the Fog index) in our sample was 1.93. The first quartile had a value of 14.7 (corresponding to the reading level of a college sophomore), median 16 (corresponding to the reading level of a college senior), and the third quartile 17.2 (corresponding to the reading level of a college graduate). In comparison, the Fog index of popular press articles is about 10, and the Fog index of academic articles is typically between 18 and 22.

¹⁰ In line with these patterns, when we re-estimated our models by replacing simplicity with these four compositional attributes, only clarity was significant and had a positive effect on opinion convergence.

of a call (see Kwon & Lee, 2003, for more explanation). The basic premise of this approach is that a document can be represented as a vector by listing the counts of specific words, and one can measure the level of similarity between them by using the angle between the two vectors representing the two documents. When two documents are very similar, the angle between the vectors representing them will be close to 0, and the cosine of that angle will be close to 1. This approach was developed in the field of natural language-processing and has been widely used in prior research to measure document similarity (see Loughran & McDonald, 2016, for a review).

To apply this approach, for each conference call, we parsed the entire transcript into two documents: the MD document and the Q&A document. We then created two vectors for the counts of all function words spoken by executives in these two documents, v_{MD} for the MD document and v_{QA} for the Q&A document:

$$v_{MD} = (\omega_1, \omega_2, \dots, \omega_n) \\ v_{QA} = (\psi_1, \psi_2, \dots, \psi_n),$$

where ω_i and ψ_i are the counts of each function word i used in these two documents, respectively, and n is the total number of unique function words used. We used the following formula to compute the cosine similarity between these two vectors, v_{MD} and v_{QA} :

$$\text{Function word similarity}_{MD, QA} = \cos(\theta) \\ = \frac{v_{MD} \cdot v_{QA}}{\|v_{MD}\| \|v_{QA}\|}$$

where $v_{MD} \cdot v_{QA}$ is the dot product of the two vectors, $\|v_{MD}\|$ is the length of the vector for the counts of function words in the MD, and $\|v_{QA}\|$ is the length of the vector for the counts of function words in the Q&A. Assume, for example, that only four function words are used in the Q&A (say, “although,” “the,” “to,” and “we”) and these words are used 5, 20, 8, and 3 times, respectively. In this case, the vector v_{QA} will be (5, 20, 8, 3) and the length of this vector would be equal to $\sqrt{(5^2 + 20^2 + 8^2 + 3^2)}$ or the square root of the sum of the squares of this vector’s constituent parts. A higher score for this measure indicates greater similarity. We then subtracted this similarity score from 1 to capture the dissimilarity in the use of function words between the MD and the Q&A portions of the calls, which we considered evidence of a more unscripted Q&A delivery by executives. Similar to our construction of content newness, we adjusted our measure of unscriptedness for document length (Lee, 2016). In our sample, unscriptedness during the Q&A varied between 4% (nearly

entirely scripted) and 44% (nearly half unscripted), with a mean of 16%.

Thus far, only Lee (2016) and the current study have used this measure of delivery unscriptedness. We checked the validity of the measure in two ways. Building on the observation that an unscripted talk is often characterized by disfluencies, such as hesitations, pauses, and fragmented sentences (Dufour et al., 2014; Lee, 2016), we first studied the occurrence of pauses (when a speaker pauses in the middle of a talk, and this unexpected interruption is identified using “—” symbol in the transcript) and fragmented sentences (i.e., sentences that begin with “and,” “but,” and “so”). After ranking our unscriptedness measure into deciles, we found that the abnormal number of pauses per sentence and the abnormal percentage of fragmented sentences were both higher in the top unscriptedness decile than in the bottom decile.¹¹ These analyses improved our confidence in the convergent validity of our measure of delivery unscriptedness. Second, we checked whether our measure of delivery unscriptedness could be differentiated from other established measures of verbal delivery. Unfortunately, there are no other verbal delivery measures we could capture using textual data. Accordingly, we checked the correlation between delivery unscriptedness and speaking rate, a verbal delivery characteristic that we were able to measure using audio data (e.g., DeGroot & Motowidlo, 1999; Krause & Braida, 2002). Because companies started to provide voice recordings of their conference calls to the public mostly after 2015 (after the observation period of our study), we manually collected a random sample of 100 voice recordings of conference calls between 2019 and 2020. We analyzed these calls for rate of speech and calculated delivery unscriptedness based on our measure. On average, executives in these conference calls spoke at a speed of 274 words per minute. The correlation between these two unrelated constructs was .019 and not significant ($p = .87$). This finding gives us some confidence regarding the divergent validity of delivery unscriptedness.

¹¹ The abnormal number of pauses per sentence was 0.146 in the top unscriptedness decile and 0.077 in the bottom decile; and the abnormal percentage of fragmented sentences was 0.125 in the top unscriptedness decile and 0.078 in the bottom decile. As an additional check, we re-estimated our models replacing unscriptedness with abnormal pauses and fragmented sentences. In these regressions, both measures had a positive and significant effect on opinion convergence with coefficients of 0.009 ($p = .067$) and 0.014 ($p = .060$), respectively.

We used standardized measures of our independent variables (namely, content newness, composition simplicity, and delivery unscriptedness) in the regressions to make their coefficients more comparable. Table 1 contains a list of existing constructs in corporate communication literature, including our independent variables, and summarizes their definition, measurement, and relation to our constructs.

Moderator: Earnings Uncertainty

Research in the accounting and finance literature has long recognized that publicly traded firms differ significantly in terms of their expected prospects (e.g., Chen & Zhang, 1998; Fama & French, 1995). Previous studies have shown that firms with a higher book-to-market ratio face greater uncertainty in their future earnings compared with other firms (e.g., Chen &

TABLE 1
Comparison of Existing Constructs in Corporate Communication Literature

Construct	Definition	Measurement Example	Illustrative Studies	Relation to Our Constructs
<i>Content</i>				
Content newness (repetition)	Content that has not been (has been) disclosed before	Similarity between the content words of previously released information and newly released information	Brown and Tucker (2011)	Our measure of content newness
Sentiment	Statements with positive (negative) emotions	Positive (negative) emotion words in LIWC dictionaries, General Inquirer dictionaries, or Loughran and McDonald's sentiment lexicon	Rhee and Fiss (2014)	Unrelated to our constructs of interest ^a
Topic discipline (diversity)	Consistency (diversity) of topics covered in a message	The proportion of contents allocated to each topic	Choudhury et al. (2019)	Unrelated to our constructs of interest
<i>Composition</i>				
Language simplicity (complexity)	Language that make the content of the communication easier to understand	The Fog index, which takes into account average sentence length and number of complex words used	Guo et al. (2020)	Our measure of composition simplicity
Language certainty (tentativeness) ^b	Language that conveys certitude regarding the likelihood of outcomes or actions	Certainty (tentative) word dictionaries in LIWC. Examples of certainty: "always," "totally." Examples of tentativeness: "perhaps," "might."	Graf-Vlachy et al. (2020)	Unrelated to our constructs of interest
Language clarity (vagueness)	Language that make communication precise in meaning and easy to paraphrase precisely	Hiller's Communication Vagueness Dictionary. Examples: "may," "nearly," "sometimes"	Guo, Yu, and Gimeno (2017)	Positively related to composition simplicity
Language concreteness (abstractedness)	Language that provide specific information and denotes tangibility	Dictionaries of verbs (e.g., run, eat), numbers (e.g., zero, dozen), and past focused words (e.g., did, ago) in LIWC	Pan et al. (2018)	Unrelated to our constructs of interest

TABLE 1
(Continued)

Construct	Definition	Measurement Example	Illustrative Studies	Relation to Our Constructs
Language activity	Language that indicates movement, action, change, and the implementation of ideas	Dictionaries of activity-oriented words in <i>Diction 7</i> . Examples: “finish,” “generate,” “overturn,” “prevent”	Seyranian and Bligh (2008)	Unrelated to our constructs of interest
Language of comparison	Language that establishes ordering among objects or makes comparisons between them	Comparative word dictionaries in LIWC. Examples: “greater,” “higher,” “best”	Graf-Vlachy et al. (2020)	Unrelated to our constructs of interest
Language of differentiation	Language that draws distinctions or describes contrasts	Differentiation word dictionaries in LIWC. Examples: “except,” “whereas,” “however,” “unlike”	Crilly et al. (2016)	Unrelated to our constructs of interest
Language vividness	Language that can easily and quickly provoke a sensory experience such as a mental image or sound	Martindale Regressive Image Dictionary of image-based and concept-based words. Examples: “sweat,” “hand,” “root,” “heart”	Emrich et al. (2001)	Unrelated to our constructs of interest
Metaphor	Figurative language that conveys thoughts and feelings by describing one domain through another domain	Manual coding based on context-specific custom dictionaries developed by researchers	König et al. (2018)	Unrelated to our constructs of interest
<i>Delivery</i> Unscriptedness (scriptedness)	Extent to which a speaker delivers a talk without relying on a script	Difference (similarity) between the use of function words between scripted and unscripted communication	Lee (2016)	Our measure of delivery unscriptedness
Speaking rate	Speed at which a person speaks	Number of words spoken per minute	DeGroot and Motowidlo (1999)	Unrelated to our constructs of interest
Speaking volume	How loudly a person speaks	Loudness of speaking in audio recordings	Holladay and Coombs (1993)	Unrelated to our constructs of interest

^aBy constructs of interest, we refer to the key independent variables in this study: content newness, composition simplicity, and delivery unscriptedness.

^bLanguage tentativeness is conceptually and methodologically identical to the language of nuance in Graf-Vlachy et al. (2020). The main difference is that Graf-Vlachy and colleagues supplemented the LIWC dictionaries with about 15 words from the Loughran and McDonald’s dictionaries of strong and weak words.

Zhang, 1998; Fama & French, 1995; Griffin & Lemmon, 2002). Accordingly, we used the book-to-market ratio as our measure of *earnings uncertainty*. Specifically, we calculated it as the total value of shareholders’ equity (i.e., the book value) at the end of the most recent quarter before the conference call and divided by the market value of outstanding shares at the end of that quarter. In our sample, the average book-to-market ratio was .63, with a standard deviation of .47. The ratio was greater than 1 in 16% of the observations. In

robustness checks, we replicated the analyses using four alternative measures of earnings uncertainty—analysts’ forecast dispersion, debt-to-equity ratio, income volatility, and return volatility—and yielded consistent results.

Control Variables

To account for alternative mechanisms that might affect the information environment of a firm and the

investor opinion convergence, we controlled for various factors related to the firm, security (stock), and conference calls. We included three firm-related control variables. We first controlled for *firm size* (measured by the logged number of employees) and *firm slack* (measured by the current ratio) at the end of the fiscal year before the conference call. Larger and more resourceful firms tend to provide greater voluntary disclosure (Petkova et al., 2012). By implication, therefore, the information obtained from the Q&A sessions will be relatively more helpful to investors in their valuations of smaller firms because less information is readily available about these firms. We also controlled for *firm performance* (measured by return on assets, ROA) at the end of the most recent quarter before the conference call. Management is typically less willing to disclose information when a firm is not performing well (Abrahamson & Park, 1994).

Among security-related variables, we first controlled for whether the earnings announcement preceding the call was an earnings surprise. Investors may react differently to negative and positive surprises. There is evidence that executives of companies that miss analysts' earnings expectations face more intense questioning during conference calls (Hollander et al., 2010). We controlled for negative and positive earnings surprises separately. We calculated the absolute value of the difference between the focal firm's reported quarterly earnings and the median of the securities analysts' consensus estimates at the end of the most recent quarter preceding the conference call and scaled it by the firm's stock price. *Negative earnings surprise* was equal to this value when the firm's actual earnings were less than consensus estimates, and 0 otherwise; *positive earnings surprise* was equal to this value when the firm's actual earnings were higher than consensus estimates, and 0 otherwise. We controlled for the *pre-announcement spread* (measured by the average amount of bid-ask spread of a firm's stock on the day before the announcement of earnings). We did this to take into account the possibility that changes in the bid-ask spread after the call might reflect a mechanical return of the spread to its pre-announcement levels after an earnings surprise. Additionally, we controlled for *return volatility* (the variability of a firm's stock price over time, as measured by the standard deviation of a firm's monthly returns over the previous 12 months). We also controlled for *share turnover* (the liquidity of a firm's stock, as measured by the average daily trading volume over the previous 12 months). Several studies have shown that,

when a firm provides a rich information environment, thereby reducing the information gap between investors, its stock tends to have lower return volatility and higher liquidity (see Healy & Palepu, 2001, for a review). We also controlled for *analyst coverage*. The availability of information from a greater number of analysts may reduce investors' need to rely on executive communication to inform their valuations. We measured analyst coverage by the natural log of the number of securities analysts covering a firm's stock at the end of the most recent quarter before the earnings conference call (Zuckerman, 1999).

Finally, we included two conference call-related variables. First, we controlled for *call length* (measured by the logged number of words in the Q&A portion of a conference call), which tends to be positively related to the amount of information provided by the Q&A (Lee, 2016). Second, we controlled for *call tenor*, the level of positive tenor expressed by executives during the conference call. To measure this variable, we first calculated the number of words used in a conference call that have an emotional tone, using the Linguistic Inquiry and Word Count (LIWC) software (Tausczik & Pennebaker, 2010). We then divided the number of positive emotion words by the number of emotional words to measure the level of positivity being expressed by the executives.¹² We controlled for call tenor to account for the possibility that investors' opinions might be influenced by the positivity/negativity of the executives' tone (Mayew & Venkatachalam, 2012).

Table 2 reports the summary statistics and correlations between these variables. In our sample, an average firm had over 14,000 employees and \$2.31 of liquid assets available to cover each dollar of its short-term debt (slack), and its ROA was about 1.5%. Its stock had low return volatility (0.12) and moderate share turnover (0.18). On average, executives' Q&A sessions contained 2,932 words (in comparison, a typical presidential inauguration speech contains 1,873 words).

¹² Although several other dictionaries also measured the use of positive and negative words (e.g., the Harvard IV-4 dictionary, DICTION, Loughran and McDonald's dictionary), we used the LIWC dictionary, because it matched the construct that we were measuring better. In supplementary analyses, we replicated the regressions by using the dictionary of positive and negative words and phrases developed by Loughran and McDonald (2011). The regressions yielded fully consistent results.

TABLE 2
Mean, Standard Deviations, and Bivariate Zero-Order Correlations

	Mean	SD	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1 Investor opinion convergence	0.329	0.857															
2 Content newness ^a	0.492	0.094	-.003														
3 Composition simplicity ^a	9.762	1.926	.037	-.005													
4 Delivery unscriptedness ^a	0.157	0.086	.018	.036	.221												
5 Earnings uncertainty	0.625	0.470	.089	-.007	.021	.025											
6 Firm size (log)	1.155	1.804	-.069	-.048	-.201	-.050	-.099										
7 Firm slack	2.342	1.600	-.021	.033	.111	.030	.022	-.362									
8 Firm performance	0.015	0.024	-.016	.002	.044	.021	-.155	.042	.057								
9 Negative earnings surprise	0.008	0.227	.037	.004	-.002	-.004	.139	-.026	-.007	-.020							
10 Positive earnings surprise	0.004	0.047	.007	.001	.015	.007	.011	-.040	.006	.293	-.003						
11 pre-announcement spread	0.413	0.463	.194	.014	-.046	-.032	-.047	.056	-.094	.005	-.001	-.012					
12 Return volatility	0.118	0.111	.012	.001	.042	.016	.084	-.140	.074	-.016	.044	.055	-.106				
13 Share turnover	0.184	0.159	-.127	.015	-.102	-.041	-.015	.043	.087	.027	.006	.007	-.123	.225			
14 Analyst coverage (log)	1.686	0.853	-.143	-.012	-.192	-.080	-.150	.523	-.158	.086	-.035	-.039	-.021	-.127	.266		
15 Call length (log)	7.796	0.753	-.059	.000	.000	-.000	-.115	.234	-.044	.030	-.017	-.030	.010	-.044	.008	.042	
16 Call tenor	0.459	0.071	-.014	.087	-.001	.021	-.003	-.010	.004	.022	-.008	.007	-.000	.004	-.006	-.060	-.021

Notes: N = 10,415. Those with an absolute value greater than .020 are significant at $p < .05$.

^aThe regression models use the standardized values of these variables.

There were only two bivariate correlations between right-hand-side variables (those between firm size and firm slack and between firm size and analyst coverage) greater than |.3|. Regression results were qualitatively unchanged when we dropped these variables from the model specification one by one and in combinations. We tested the impact threshold of a confounding variable (ITCV) to assess how strongly correlated an omitted variable would have to be to overturn the results of our analyses (Busenbark, Gamache, Yoon, & Withers, 2021; Frank, 2000). ITCV scores for composition simplicity and delivery unscriptedness were .18 and .25, respectively. These scores imply that, to overturn the results, the correlation between simplicity and investor opinion convergence with an omitted confounding variable would have to be about .43 ($= \sqrt{.18}$); similarly, the correlation between unscriptedness and investor opinion convergence with an omitted confounding variable would have to be about .50 ($= \sqrt{.25}$). We also calculated ITCV scores for the interaction terms of our independent variables with earnings uncertainty. This yielded impact thresholds of .21 ($= \sqrt{.04}$) for the interaction term with newness; .53 ($= \sqrt{.28}$) for the interaction term with simplicity; and .49 ($= \sqrt{.24}$) for the interaction term with unscriptedness. To put these values in perspective, .20 was the highest correlation between our control variables and any of our dependent and independent variables. Hence, the ITCV tests (with the exception of content newness, which we will discuss below) indicated that overturning the results would require an omitted variable with an impact larger than the strongest control variable in our model specification.

However, the ITCV score for content newness was much lower (.001), implying that, to overturn the results, the correlation between newness and investor opinion convergence with an omitted confounding variable would have to be about .025 ($= \sqrt{.001}$). Accordingly, we followed Busenbark and colleagues' (2021: 34) recommendation and ran a two-stage model using general counsel (a dummy variable that equals 1 if the general counsel is among the top-five highest paid executives in a firm in a given year, 0 otherwise) as an instrument for content newness. It has been increasingly recognized that, to minimize the risk of litigation and regulatory penalties, general counsels play an important monitoring role in firms' disclosure activities, especially when they are on the top management team (DeMott, 2005; Kwak, Ro, & Suk, 2012). Thus, we expect that the presence of a general counsel on a firm's top

management team would reduce the disclosure of new content in the Q&A session. As expected, we found that general counsel ($p < .01$) had a negative and significant relationship with content newness in the first-stage regression. In the second stage, the predicted value of content newness still had a positive and significant relationship with investor opinion convergence ($p < .05$), alleviating the concerns of potential omitted variable bias.

Estimation

To test our hypotheses, we ran regressions with speaker and year-quarter fixed effects. Speaker fixed-effects account for executives' distinct styles of communication and other time-invariant characteristics; year-quarter fixed-effects account for unobserved heterogeneity across time periods. For each conference call, we identified all of the executives who participated in the call and identified the executive who spoke the most words during the call as the spokesperson of that call (Lee, 2016). To alleviate any concerns for serial correlation in our models, we clustered the standard errors by firm (i.e., we let observations be independent across firms, but not necessarily across conference calls by the same firm) (Petersen, 2009).

RESULTS

We hypothesized that the levels of content newness (Hypothesis 1), composition simplicity (Hypothesis 2), and delivery unscriptedness (Hypothesis 3) of executives' verbal responses in the Q&A portion of conference calls would be positively associated with investor opinion convergence. The regression results reported in Table 3 support these hypotheses, because the coefficients of content newness ($p < .05$), composition simplicity ($p < .05$), and delivery unscriptedness ($p < .01$) were positive and statistically significant, as predicted (Model 2).

Regarding economic significance, holding all other variables at their mean values, the convergence of investors' opinions was 11.5% higher for Q&A sessions with high levels of content newness (one standard deviation above the sample mean) compared with Q&A sessions with low levels of content newness (one standard deviation below the sample mean); 14.0% higher for Q&A sessions with high levels of simplicity, compared with Q&A sessions with low levels of simplicity; and 18.2% higher for Q&A sessions with high levels of unscriptedness, compared with Q&A sessions with low levels of

unscriptedness. For an average firm in our sample (with a pre-call spread of \$.216 per share), these percentages corresponded to a difference of \$.026, \$.031, and \$.040 per share, respectively.

Hypothesis 4 predicted that the positive effect of executive communication (namely, newness of content, simplicity of composition, and unscriptedness of delivery) on investor opinion convergence would be amplified for those firms with high earnings uncertainty. The regression results (Models 3 to 5) support this prediction, because the coefficient of the interaction term between content newness and earnings uncertainty is both positive and significant ($p < .05$), and so are the interaction terms between content simplicity and earnings uncertainty ($p < .01$) and between delivery unscriptedness and earnings uncertainty ($p < .01$). We illustrate these interactions in Figures 2, 3, and 4. Defining the "low" level of a measure as one standard deviation below the sample mean and the "high" level as one standard deviation above the sample mean, these interactions imply that, at low levels of firms' earnings uncertainty, the convergence of investors' opinions is 7.6% (5.8%, 4.7%) higher for Q&A sessions with high levels of content newness (composition simplicity, delivery unscriptedness), compared with Q&A sessions with low levels of newness (simplicity, unscriptedness). At high levels of firms' earnings uncertainty, this difference increases to 15.7% (24.2% and 21.4% for composition simplicity and delivery unscriptedness, respectively). As we hypothesized, therefore, executive communication has a stronger effect on investor opinion convergence when a firm has high future earnings uncertainty.

We also conducted simple slope tests to determine whether the slope of the interaction lines in Figure 2–4 were significantly different. We found that, when earnings uncertainty is high, all three independent variables are significant predictors of investor opinion convergence (coefficient of content newness = .19, $p < .05$; coefficient of composition simplicity = .08, $p < .001$; coefficient of delivery unscriptedness = .06, $p < .001$). However, under conditions of low earnings uncertainty, their impacts on investors' opinion divergence varied. Although the effects of composition simplicity and delivery unscriptedness were marginally significant ($p < .10$), the effect of content newness was not significant. These findings are consistent with our observation that the impact of executive communication is significantly weaker under conditions of low earnings uncertainty.

Also, one should note that several control variables affect the change in investor opinion

TABLE 3
Regressions Predicting Investor Opinion Convergence

	(1)	(2)	(3)	(4)	(5)	(6)
Content newness		.115*	.114*	.113 [†]	.115 [†]	.113 [†]
		(.057)	(.057)	(.063)	(.063)	(.063)
Composition simplicity		.017*	.016*	.016*	.017*	.016*
		(.008)	(.008)	(.008)	(.008)	(.008)
Delivery unscriptedness		.022**	.022**	.021**	.019*	.019*
		(.008)	(.008)	(.008)	(.008)	(.008)
Content newness × Earnings uncertainty			.268*			.214
			(.132)			(.145)
Composition simplicity × Earnings uncertainty				.030**		.021*
				(.009)		(.010)
Delivery unscriptedness × Earnings uncertainty					.027**	.023**
					(.007)	(.007)
<i>Controls</i>						
Earnings uncertainty	.043**	.043**	.044**	.038**	.045**	.042**
	(.009)	(.009)	(.009)	(.009)	(.009)	(.009)
Firm size	-.037**	-.034**	-.034**	-.034**	-.035**	-.035**
	(.005)	(.005)	(.005)	(.005)	(.005)	(.005)
Firm slack	-.018**	-.019**	-.019**	-.018**	-.019**	-.018**
	(.005)	(.005)	(.005)	(.005)	(.005)	(.005)
Firm performance	-.363	-.418	-.401	-.445	-.378	-.396
	(.326)	(.326)	(.327)	(.326)	(.326)	(.327)
Negative earnings surprise	.067**	.068**	.066**	.066**	.070**	.070**
	(.023)	(.023)	(.023)	(.023)	(.023)	(.023)
Positive earnings surprise	.122	.134	.123	.154	.117	.128
	(.147)	(.147)	(.147)	(.147)	(.147)	(.148)
Pre-announcement spread	.004**	.004**	.004**	.004**	.004**	.004**
	(.000)	(.000)	(.000)	(.000)	(.000)	(.000)
Return volatility	.134*	.139*	.139*	.137*	.140*	.139*
	(.057)	(.057)	(.057)	(.057)	(.057)	(.057)
Share turnover	-.466**	-.459**	-.459**	-.454**	-.453**	-.451**
	(.046)	(.046)	(.046)	(.046)	(.046)	(.046)
Analyst coverage	-.128**	-.119**	-.119**	-.119**	-.119**	-.119**
	(.011)	(.011)	(.011)	(.011)	(.011)	(.011)
Call length	.006	-.007	-.007	-.006	-.006	-.006
	(.008)	(.010)	(.010)	(.010)	(.010)	(.010)
Call tenor	-.089	-.116	-.116	-.117	-.115	-.115
	(.088)	(.088)	(.088)	(.088)	(.088)	(.088)
<i>Fixed-effects</i>						
Speaker	Included	Included	Included	Included	Included	Included
Year-quarter	Included	Included	Included	Included	Included	Included
Log-likelihood	-8311.12	-8302.07	-8301.73	-8296.68	-8293.50	-8291.06
Change in log-likelihood		9.05	9.39	14.44	17.62	20.06

Notes: $N = 10,415$. Robust standard errors, clustered by firm, in parentheses. Constant included in all models. Two-tailed tests.

[†] $p < .10$

* $p < .05$

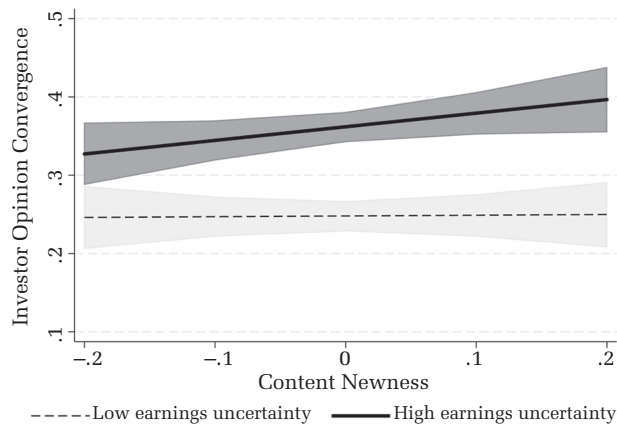
** $p < .01$

convergence after a conference call. The direct effects of earnings uncertainty and return volatility are statistically significant and indeed echo our discussions related to Hypothesis 4. The coefficients for firm size and firm slack were negative and statistically significant, consistent with our expectations. Moreover, the coefficients for negative earnings surprise were positive and significant, but the

coefficients for positive earnings surprise were positive but not statistically significant. This is consistent with prior findings that investors tend to react more strongly to bad news. In line with the prior research, we found pre-announcement spread is positively associated with investor opinion convergence, implying that, when investors face higher information asymmetry, the effect of conference

FIGURE 2

The Interaction Effect between the Level of Content Newness and Earnings Uncertainty on Investor Opinion Convergence



calls is more significant. Lastly, as expected, share turnover and analyst coverage both had negative and significant influences on investor opinion convergence.

Interdependence of Content Newness, Composition Simplicity, and Delivery Unscriptedness

Although we hypothesized for the independent effects of newness, simplicity, and unscriptedness, their effects could be interdependent. To check this

FIGURE 3

The Interaction Effect between the Level of Composition Simplicity and Earnings Uncertainty on Investor Opinion Convergence

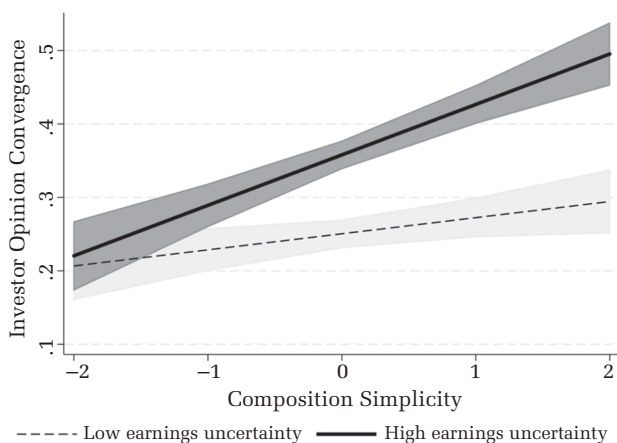


FIGURE 4

The Interaction Effect between the Level of Delivery Unscriptedness, and Earnings Uncertainty on Investor Opinion Convergence



possibility, we ran supplementary analyses, including pairs of interactions between our independent variables as well as a three-way interaction among them (see Table 4). In these regressions, the interaction term between newness and simplicity and the interaction term between newness and unscriptedness were not significant, with very high p values (.526 and .877, respectively). In contrast, the interaction term between simplicity and unscriptedness was positive and highly significant ($p = .004$). Finally, the three-way interaction term among these three aspects of executive communication was negative and significant at the 10% level ($p = .088$).

These results suggest two main inferences. First, executive communication increases the convergence of investors' opinions even more when such communication is conveyed using simple language *and* in an unscripted fashion. In other words, content simplicity and delivery unscriptedness are complements in influencing investor opinion convergence. This resonates with our prediction that the effectiveness of executive communication as sense-giving activities hinges on the properties of the information being conveyed, especially its perceived comprehensibility and credibility. Second, the influence of content newness on investor opinion convergence is independent of the level of composition simplicity and delivery unscriptedness. The negative sign of the three-way interaction term implies that the interactive effect of simplicity and unscriptedness is somewhat lower when content is newer. Although our data do not allow us to disentangle the specific

TABLE 4
Regression Results for Two-way and Three-way Interactions among Content Newness, Composition Simplicity, and Delivery Unscriptedness (Abbreviated Results)

	(1)	(2)	(3)
Content newness	.115* (.057)	.115 ⁺ (.063)	.145* (.066)
Composition simplicity	.017* (.008)	.021* (.009)	.021* (.009)
Delivery unscriptedness	.022** (.008)	.011 (.009)	.011 (.009)
Content newness × Composition simplicity		−.002 (.006)	−.004 (.007)
Content newness × Delivery unscriptedness		−.005 (.007)	.001 (.008)
Composition simplicity × Delivery unscriptedness		.022** (.008)	.021** (.008)
Content newness × Composition simplicity × Delivery unscriptedness			−.013 ⁺ (.008)
Controls	Included	Included	Included
Speaker fixed-effects	Included	Included	Included
Year-quarter fixed-effects	Included	Included	Included
Log-likelihood	−8302.07	−8279.56	−8296.11
Change in log-likelihood		4.51	5.96

Notes: $N = 10,145$. Robust standard errors, clustered by firm, in parentheses. Constant included in all models. Two-tailed tests.

⁺ $p < .10$

* $p < .05$

** $p < .01$

mechanisms that link newness to investor opinion convergence, we suspect these findings indicate that new content influences investor opinion convergence mainly through the resulting reduction in the information gap between diversely informed investors. If this is the case, the sense-giving role of executive communication (captured by simplicity and unscriptedness) is contingent not on whether new information is revealed, but on whether investors are open to be influenced in an ambiguous situation.

Robustness Tests and Supplementary Analyses

We conducted numerous additional analyses to assess the robustness of the results to the sample construction, model specification, and measurement of our independent variables. First, we examined the representativeness of our sample. By studying calls held between 11 a.m. and 2 p.m. EST (8 a.m. and 11 a.m. Pacific Standard Time), we missed many firms located on the West Coast because firms rarely hold conference calls that early in the day. To check for potential sample-selection bias, we performed a two-stage Heckman model. In the first-stage regression, we included a dummy variable for firms whose headquarters are located in the West Coast as an additional

predictor of the likelihood that a call would be held during trading hours. This variable was statistically significant ($p < .001$), suggesting that the exclusion restriction was satisfied. In the second-stage regression, the lambda value was not statistically significant ($p = .73$), implying that selection is unlikely to be a major concern that influences the reported results (which did remain unchanged).¹³

Second, we estimated our regressions by including speaker fixed-effects, which only control for the time-invariant characteristics of the speaker in the model specification. In supplementary analyses, we checked whether the results were sensitive to the

¹³ Examining the full data set, we also found that 38.2% of firms always hold their conference calls during trading hours, 40.2% of firms always hold their calls after trading hours, 16.8% switched their call times only once during the seven-year period of our study, and 4.8% switched their call time multiple times. Hence, it seems that the timing of conference calls is more closely related to the policy of a company than to any identifiable strategic reasoning. Furthermore, these four groups of firms did not differ in any observable characteristic (except for size—those firms that regularly held conference calls during trading hours were generally larger).

inclusion of time-variant speaker characteristics. We were able to obtain data for two such characteristics: speaker age (logged number of years since birth) and tenure (logged number of years since joining the company). These two variables, as expected, were positively correlated (.41). When included in the regressions, neither was significant, and the main results remained unchanged. We did not include these variables in our model specification because we were unable to obtain the corresponding data from ExecuComp and BoardEx for more than one-fifth of our working sample because of inconsistencies in name formats.

Third, we ran regressions to account for any potential endogeneity of our unscriptedness measure by using shareholder lawsuits in an industry (measured by the number of security-related class action lawsuits in the year before the conference calls) as an instrument. One might argue that executives in industries with higher risks of shareholder litigation are less likely to communicate with their investors in an unscripted way because the risk of shareholder litigation can reduce managers' incentive to disclose information, especially voluntary disclosures (Rogers & Van Buskirk, 2009). Using data from Stanford Law School's Securities Class Action Clearinghouse, we calculated the number of security-related class action lawsuits in an industry (i.e., four-digit SIC code) in the year before conference calls. We found, as expected, that shareholder lawsuits ($p < .01$) represent a negative and significant predictor of the level of delivery unscriptedness in the first-stage regression. In the second-stage regression, as expected, the predicted value of delivery unscriptedness was positively associated with investor opinion convergence ($p < .05$).

Fourth, we reproduced Models 3–6 by replacing our measure of earnings uncertainty (book-to-market ratio) with four alternative measures and having them interact with the independent variables. These four measures were the analysts' forecast dispersion (measured by standard deviation of the analysts' forecasts in the 12 months before the conference call) (Johnson, 2004), debt-to-equity ratio (measured by the ratio of total liabilities to shareholder equity) (Opler & Titman, 1994), income volatility (measured by the standard deviation of ROA in the five years before the conference call) (Miller & Bromiley, 1990), and return volatility (measured as noted above). In these regressions, the coefficient of the interaction term with content newness was significant only when return volatility was used, whereas the coefficients of the interaction terms with both

simplicity and unscriptedness remained positive and significant, regardless of which alternative measure was used. Thus, consistent with Hypothesis 4, these results show that earnings uncertainty is a critical moderator that amplifies the positive relationship between executive communication and investor opinion convergence.

Our final two tests attempted to further unpack the underlying mechanisms of investors' demand for information. In our fifth supplementary analysis, we checked whether there is firm-level heterogeneity in our theoretical model based on the investors' assessment of the firm before the calls. Previous studies have argued that highly regarded firms tend to suffer less from negative events (e.g., corporate crises), also known as "the reservoir of goodwill hypothesis" (e.g., Jones, Jones, & Little, 2000; Yu, Sengul, & Lester, 2008). To check, we re-estimated the regressions, including the level of a firm's prominence (measured by the firm's average stock returns in the 12 months before the conference call and adjusted for average market returns) and its interaction with our independent variables. We found that the effects of content newness and composition simplicity on investor opinion convergence are weaker when firms have reported better stock performance in the past 12 months; however, good/bad performance does not seem to influence the effect of delivery unscriptedness. Thus, our finding is consistent with previous studies, in that executive communication is much more consequential in shaping investors' opinions for those firms that have performance problems (and hence their investors' demands for information is high) than for firms that perform well.

Similarly, if executive communication increases the perceived comprehensibility and credibility of the information conveyed, one may expect such communications to be even more influential in shaping investors' opinions during periods of high uncertainty and turbulence (when information is usually scarcer, unreliable, and difficult to obtain). To verify this possibility, we reran the analyses, restricting our sample to the period surrounding the collapse of Lehman Brothers in September 2008. Accordingly, we identified the pre-crisis period as the two years between September 2005 and September 2007, and the crisis period as the two years between September 2008 and September 2010.¹⁴ Consistent with our expectations, the effects of

¹⁴ We excluded the immediate 12 months before Lehman's collapse from the pre-crisis period to rule out any potential anticipatory effects and uncertainty related to the exact start date of the financial crisis. In contrast,

newness, simplicity, and unscriptedness were more pronounced during the crisis period.

DISCUSSION

The primary objective of this study was to understand how executive verbal communication influences the perceived comprehensibility and the credibility of information conveyed, which in turn shapes investor opinion convergence. Focusing on three essential aspects of verbal communication (i.e., content, composition, and delivery), we found that content newness, composition simplicity, and delivery unscriptedness are all positively associated with investor opinion convergence. Moreover, to explore the boundary conditions that govern these relationships, we focused on the role of earnings uncertainty. We showed that the relationships between the newness, simplicity, and unscriptedness of executive verbal communication and investor opinion convergence are strengthened when a firm's earnings uncertainty is higher.

This research extends the literature on strategic leadership and corporate communication in several meaningful ways. Research on corporate communication began in the 1980s, with studies that largely relied on manual reading to examine the content of executives' written communication to shareholders (e.g., Bettman & Weitz, 1983; Salancik & Meindl, 1984). Recently, this literature has begun to examine how the content and composition of executive verbal communication (i.e., subtle linguistic attributes, such as concrete and metaphorical language) influence the valuations of investors and securities analysts (König et al., 2018; Pan et al., 2018; Rhee & Fiss, 2014). We joined these studies and extended them by adding delivery as a third and important aspect of verbal communication. This extension is a natural and much-needed one, given that content, composition, and delivery are the three fundamental elements of a message (Den Hartog & Verburg, 1997). Moreover, an identical message can have varying effects on its recipients, depending on how it is delivered. Thus, studies about how corporate communication influences investors would be incomplete without considering the role of delivery. In this regard, our study offers a comprehensive study of the extent to which various aspects of executive verbal communication affect investors' valuations.

Additionally, we used various natural language processing techniques (e.g., text embedding, cosine similarity, and topic modeling) to analyze executives' communications. This represents an early attempt at using such sophisticated linguistic approaches (compared with the word-count approach largely used in the existing research) in the strategic management literature.

Second, we added a new theoretical lens to the literature on corporate communication. Most studies within this body of work have adopted an impression-management perspective to theorize how skillfully crafted language can be used as an effective tactic to create or maintain a favorable organizational image in the eyes of investors (as reflected by stock price changes) (e.g., Pan et al., 2018; Whittington et al., 2016). Our perspective complements the existing studies by highlighting the sense-giving role of executive communication, which focuses more on information value than on the impression-management value of such communication. After all, the primary purpose of corporate communication is to help investors make sense of an ambiguous situation. Executive communication can reduce the information gap between investors by increasing the perceived comprehensibility and credibility of the information being conveyed. Hence, we believe it is important to consider both the impression management and information value of corporate communication. Delivery unscriptedness, for example, has both information value, as we theorized, and impression-management value. Talking without a script increases the risk of making a mistake or delivering an unpolished talk, and these undesirable outcomes may make executives seem unprepared and incompetent, thereby damaging investors' impressions.

Third, we investigated a new and important aspect of investor valuation that has received scant attention in the strategy and management literature: investor opinion convergence. Although the question of how executives affect investors' decisions has been widely studied by management scholars, the majority of prior studies have focused on one aspect of investor valuations: changes in stock prices (see Higgins & Gulati, 2006, as an exception). In this study, we introduced the concept of investor opinion convergence to the strategy and management literatures. Compared with the changes in stock prices, changes in the differences of opinion among investors can better depict investors' sensitivity to firms' information environment and investors' reactions to corporate communication strategies. In this regard,

our findings complement the prior research by offering a more complete picture of the effect of corporate communications on investors. Moreover, existing research on opinion convergence has taken a rational view that emphasizes the role of executive communication in supplying more information to investors (e.g., Blankespoor et al., 2014; Lee, 2016). In contrast, we offer a complementary, cognitive perspective that delves into the sense-giving role of executive communication in making information provided more comprehensible and credible.

Lastly, we introduced a novel intraday approach to study the impact of corporate communication. Although numerous previous studies have demonstrated the advantages of this approach (e.g., Chen et al., 2018; Matsumoto et al., 2011), the complexities and difficulties associated with processing the massive amount of intraday trading data creates challenges for researchers. To the best of our knowledge, this is the first study in the strategy and management field that uses an intraday approach to capture investors' immediate reactions to corporate communication. Given the increased prevalence of high-speed trading, studying investors' reactions at such a fine-grained level will help researchers gain more accurate insights about when and how investors react to critical corporate events. More important, by using our novel research design, we were able to parse the effects of earnings announcements and achieve better control of the noise that occurs between the earnings announcement and executive communication. In other words, our novel research design has let us more cleanly and precisely capture the power of language.

Future Research Directions

This study highlights numerous promising avenues for future research. Constrained by data availability, we used transcripts of conference calls to focus on unscriptedness, the only viable delivery attribute that can be analyzed. One promising area for future research is to explore how other aspects of executives' verbal (e.g., voice volume and pitch, rate of speaking, the use of pauses and inflection, and pronunciation) and nonverbal (e.g., facial expression, eye contact, and hand gestures) delivery may influence investors' decisions and valuations. All these other aspects of delivery may also carry important information cues for investors to make sense of executive communication.

Another promising avenue for future research is exploring whether and how executive

communication shapes the perceptions of various audience groups. In this study, we focused on two audience groups: the investors and, in the supplementary analyses, the analysts. Nevertheless, there are other external constituents that firms rely on for resources and support, such as journalists and regulators (e.g., Pollock & Rindova, 2003). Because the demands of these important stakeholders tend to be different (König et al., 2018), they likely look for different types of information. Hence, exploring the link between executive verbal communication and the opinions of different audience groups may uncover important heterogeneities regarding the effects of such communication on stakeholders' perceptions and evaluations.

Finally, there is value in exploring alternative sources and measures of investor opinion convergence. We followed the prior research and used the bid-ask spread, one of the most widely used proxies for studying differences in investors' opinions. Nevertheless, it is only a proxy for the aggregated level of agreement among investors. We hope our study will inspire more research that will use other research methods, such as experiments or simulations (e.g., Awamleh & Gardner, 1999; Crilly, 2017). A particularly promising avenue in our view rests on direct measures of the opinions of individual investors. Technological development and the evolution of social media, such as Twitter, have made such measures available and possible. There are already creative uses of these kinds of data sources in studies of corporate disclosure (e.g., Blankespoor et al., 2014; Giannini, Irvine, & Shu, 2019). By measuring the valuations of individual investors and linking them to demographic and other characteristics of investors, future studies can go beyond aggregated measures and more closely analyze the antecedents and consequences of investor opinion convergence.

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